

tf.compat.v1.linalg.diag Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/linalg/diag

Returns a batched diagonal tensor with given batched diagonal values.

Function Signatures

```
tf.compat.v1.linalg.diag( diagonal, name='diag', k=0,  
num_rows=-1, num_cols=-1, padding_value=0, align='RIGHT_LEFT'  
)
```

Code Examples

```
tf.compat.v1.linalg.diag(  
diagonal,  
name='diag',  
k=0,  
num_rows=-1,  
num_cols=-1,  
padding_value=0,  
align='RIGHT_LEFT'  
)
```

```
tf.compat.v1.linalg.diag(
    diagonal,
    name='diag',
    k=0,
    num_rows=-1,
    num_cols=-1,
    padding_value=0,
    align='RIGHT_LEFT'
)
```

```
output[i, j, ..., l, m, n]
= diagonal[i, j, ..., l, n-max(d_upper, 0)] ; if n - m == d_upper
padding_value                               ; otherwise
```

```
output[i, j, ..., l, m, n]
= diagonal[i, j, ..., l, n-max(d_upper, 0)] ; if n - m == d_upper
padding_value                               ; otherwise
```

```
output[i, j, ..., l, m, n]
= diagonal[i, j, ..., l, diag_index, index_in_diag] ; if k[0] <= d
<= k[1]
padding_value                                     ; otherwise
```

```
output[i, j, ..., l, m, n]
= diagonal[i, j, ..., l, diag_index, index_in_diag] ; if k[0] <= d
<= k[1]
padding_value                                     ; otherwise
```

```
offset = max_diag_len - diag_len(d) ; if (`align` in {RIGHT_LEFT,
RIGHT_RIGHT}
and `d >= 0`) or
(`align` in {LEFT_RIGHT, RIGHT_RIGHT}
and `d <= 0`)
0                                     ; otherwise
```

```

offset = max_diag_len - diag_len(d) ; if (`align` in {RIGHT_LEFT,
RIGHT_RIGHT}
and `d >= 0`) or
(`align` in {LEFT_RIGHT, RIGHT_RIGHT}
and `d <= 0`)
0 ; otherwise

```

Documentation

Returns a batched diagonal tensor with given batched diagonal values.

Returns a tensor with the contents in diagonal as $k[0]$ -th to $k[1]$ -th diagonals of a matrix, with everything else padded with padding. `num_rows` and `num_cols` specify the dimension of the innermost matrix of the output. If both are not specified, the op assumes the innermost matrix is square and infers its size from `k` and the innermost dimension of diagonal. If only one of them is specified, the op assumes the unspecified value is the smallest possible based on other criteria.

Let diagonal have r dimensions $[I, J, \dots, L, M, N]$. The output tensor has rank $r+1$ with shape $[I, J, \dots, L, M, \text{num_rows}, \text{num_cols}]$ when only one diagonal is given (k is an integer or $k[0] == k[1]$). Otherwise, it has rank r with shape $[I, J, \dots, L, \text{num_rows}, \text{num_cols}]$.

The second innermost dimension of diagonal has double meaning. When k is scalar or $k[0] == k[1]$, M is part of the batch size $[I, J, \dots, M]$, and the output tensor is:

Otherwise, M is treated as the number of diagonals for the matrix in the same batch ($M = k[1] - k[0] + 1$), and the output tensor is:

where $d = n - m$, $\text{diag_index} = k[1] - d$, and
 $\text{index_in_diag} = n - \max(d, 0) + \text{offset}$.

offset is zero except when the alignment of the diagonal is to the right.

where $\text{diag_len}(d) = \min(\text{cols} - \max(d, 0), \text{rows} + \min(d, 0))$.

For example:

Returns